

Jannatul Ferdeous

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EDUCATION

- Green University of Bangladesh**
Bachelor of Science in Computer Science and Engineering; CGPA: 3.98/4.00 2020 - 2024

SKILLS SUMMARY

- Languages:** Python, C, Java, SQL, HTML, CSS, JavaScript
- Backend:** Django, FastAPI, Flask, Celery
- Libraries:** PyTorch, TensorFlow, LangChain, LangGraph, NumPy, Pandas, OpenCV, Matplotlib, Scikit-learn
- Databases:** MySQL, SQLite, ChromaDB, Pinecone, FAISS, Redis, PostgreSQL
- Toolchain:** Docker, Git, Bitbucket, N8N, Postman, LaTeX, MS Office, VS Code, MATLAB, Cisco Packet Tracer

EXPERIENCE

- StudyNet Pty Ltd.**
AI Engineer Dec 2025 - Present
 - Driving Agentic AI initiatives** in a fast-paced, dynamic environment at an **Australian-based multinational EdTech organization**, with a strong focus on cross-functional collaboration (Onshore/Offshore) to build scalable AI solutions.
 - Architected and deployed [StudyNet.AI](#)**, a **multi-agent conversational AI platform** fully integrated with [StudyNet.io](#), leveraging LLM-powered agents, RAG pipelines, and real-time data integration to deliver end-to-end guidance on **university admissions, course selection, scholarships, visas, and migration processes** for students targeting Australia.
 - Implemented a hybrid RAG agent** for internal knowledge sources (policy documents, SOPs), utilizing vector DBs (FAISS and Pinecone) with reranking strategies to improve **retrieval accuracy by 30% and reduce latency**.
 - Designed and developed an API-first OCR-driven enquiry processing pipeline** to extract and structure data from **CVs, passports, and educational transcripts**, enabling seamless integration with AI-driven systems.
 - Integrated Azure OpenAI (GPT) APIs with parallel request orchestration** to enable scalable, low-latency processing for multi-agent workflows and real-time query handling.
 - Developed autonomous and domain-specific recommendation agents** to fetch and validate real-time data from trusted Australian sources and deliver **accurate course and scholarship recommendations**, including fees, intake periods, and personalized opportunities.
 - Deployed a Redis-based semantic caching layer with contextualization**, optimizing system in terms of **response time for high-frequency queries**.
 - Engineered memory-driven contextualization mechanisms** allowing agents to leverage **user history and prior interactions** for personalized and coherent responses.
- Green Computing and Communication Lab**
AI/ML Researcher May 2023 - Present
 - Published research in Q1 journals and conferences** on asynchronous model training in low-resource environments for industrial applications, contributing to efficient distributed learning strategies.
 - Enhanced model performance on real-world datasets** in distributed settings such as **Federated and SplitFed Learning**, improving robustness across decentralized data sources.
 - Collaborated with and mentored fellow researchers** by guiding experimentation, refining methodologies, and supporting end-to-end research and implementation workflows.
- Robo Tech Valley**
ML Engineer May 2024 - Aug 2024
 - Developed scalable and efficient data pipelines** using the **Roboflow** platform for data augmentation, preprocessing (normalization, denoising), and streamlined data annotation workflows.
 - Optimized real-time object detection performance** by improving YOLO model inference speed using TensorRT-based GPU parallelization for low-latency deployment.
 - Collaborated with cross-functional teams** to align model development with system requirements, optimize deployment pipelines, and deliver robust and production-ready object detection solutions.

PROJECTS

- **Multimodal RAG for PDF Q&A (NLP, Multimedia Processing):** A multimodal RAG pipeline for PDFs integrating OCR-based text extraction, VLMs for image understanding, and an LLM-powered Q&A chain, supported by Pinecone vector database. Tech: Python, Django, TesseractOCR, LangChain, Gemini LLM & VLM, PineconeDB, Streamlit. [GitHub:] github.com/doc-rag-ai
- **Agentic AI for Financial Document Compliance (NLP, Document Parsing):** An agent-based system that automates document compliance by extracting, validating, and cross-checking key fields across various document types. Tech: Python, LangGraph, EasyOCR, NLP Library (HuggingFace), Local LLMs (Ollama/LLaMA). [GitHub:] github.com/doc-compliance-ai
- **Real-time Bangladeshi Vehicle License-Plate Detection and Tracking (Computer Vision):** A real-time system for detection, recognition, and tracking of Bangladeshi vehicle license plates. Tech: Python, OpenCV, YOLO, EasyOCR. [GitHub:] github.com/lpr-bangladesh
- **Multimodal Automated CAPTCHA Solver (Distributed Computing, Multimedia Processing):** A multimodal CAPTCHA solver that decodes audio files and extracts text from images for automated CAPTCHA processing. Tech: Python, Whisper, EasyOCR, OpenCV, PyTorch. [GitHub:] github.com/captcha-solver-ai

PUBLICATIONS

- (Journal Q1): **IoT-Enabled Smart Workplaces with Autonomous LLM-Powered AI Agents**, Jannatul Ferdeous, Himel Saha, *et al.*, 2026. (Ongoing)
- (Journal Q1): **Device and Data Heterogeneity Aware SplitFed Learning for Digital Twin Empowered Industrial IoT**, Himel Saha, Md Nur Ahmed, Palash Roy, *et al.*, *Computer Networks (Elsevier)*, 2025. (Q1, IF: 4.6). Link: doi.org/10.1016/j.comnet.2025.111478
- (Conference): **ASFL – An Adaptive SplitFed Learning Framework for IIoT**, Himel Saha, Md Nur Ahmed, Palash Roy, Md. Abdur Razzaque, *IEEE Computer Society Bangladesh Chapter Symposium (IEEECSBDCS)*, 2024. Link: researchgate.net/publication/393446915

HONORS AND AWARDS

- **Innovation & Excellence Award, StudyNet Annual Award Ceremony, 2026** for developing the agentic AI-powered educational intelligence system (**StudyNet.AI**).
- **Top Finalist, BUET CSE Festival 2024** for presenting the research “*Adaptive SplitFed Learning Framework for IIoT*”.
- **Best Poster Paper Award, GUB CSE Carnival 2024** for the research titled “*ASFL: An Adaptive SplitFed Learning Framework for Industrial Internet of Things*”.

REFERENCES

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